Ionic and Covalent Bonds

Answer key with worked solutions

A. Understanding

1. Ionic vs covalent.

> **Ionic:** electrons are transferred from one atom (usually a metal) to another (usually a non-metal), and the oppositely charged ions attract. **Covalent:** two atoms (usually non-metals) share one or more pairs of electrons.

2. Electronegativity.

> How strongly an atom attracts the shared pair of electrons in a covalent bond.

B. Classifying bonds

3. Classify.

> (a) Cl–Cl: 0.00, **nonpolar covalent** (b) C–Cl: 3.16 − 2.55 = 0.61, **polar covalent** (c) N–H: 3.04 − 2.20 = 0.84, **polar covalent** (d) Mg–O: 3.44 − 1.31 = 2.13, **ionic** (e) Li–F: 3.98 − 0.98 = 3.00, **ionic**

4. Partial negative charge.

> C–Cl: **Cl** is δ−. N–H: **N** is δ−.

5. Increasing polarity.

> ΔEN: C–H 0.35, H–Br 0.76, O–H 1.24. Order: **C–H (least polar), H–Br, O–H (most polar)**

C. Ionic formulas

6. Formulas.

> (a) K^+ and O^{2-}: K_2O (b) Ca^{2+} and Br^-: $CaBr_2$ (c) Al^{3+} and N^{3-}: AlN (d) Mg^{2+} and P^{3-}: Mg_3P_2

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7. Ion formation.

> $\mathrm{Mg} \longrightarrow \mathrm{Mg^{2+}} + 2\,\mathrm{e^-}$ and $\mathrm{Cl_2} + 2\,\mathrm{e^-} \longrightarrow 2\,\mathrm{Cl^-}$. Overall the compound is $\mathrm{MgCl_2}$.

D. Properties

8. Melting points.

> NaCl is an ionic lattice: melting requires overcoming many **strong attractions between oppositely charged ions**, which takes a lot of energy. Methane is made of separate molecules: melting only overcomes the **weak attractions between molecules**. The covalent C–H bonds do not break.

E. Challenge

9. Conductivity.

> Conduction needs **charged particles that can move**. In the solid, the ions are held in fixed positions in the lattice. When molten or dissolved, the ions are free to move, so they can carry charge.

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